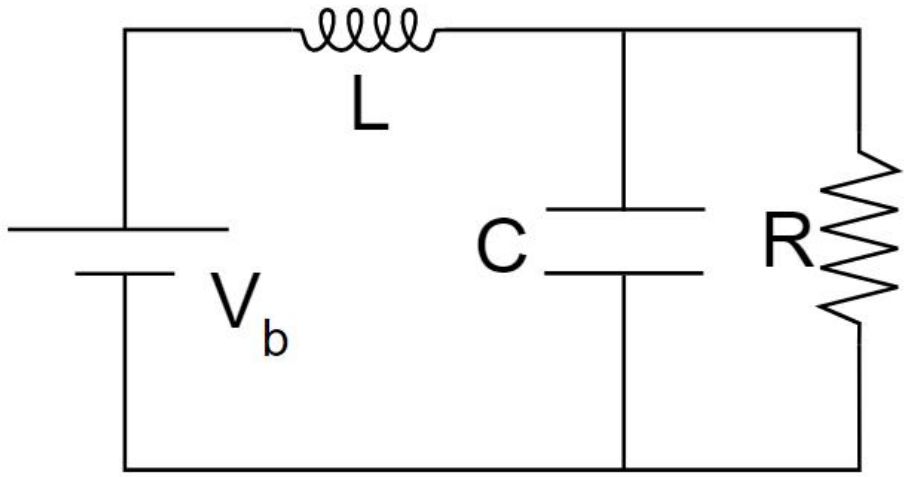
Final project

By: Yiqing Wang
Student ID:400336382

My final project is to modeling a Low-Pass RLC Filter.
It's circurt structure is shown below:



A Low-Pass RLC is a filter that made by an electrical circuit consisting of a resistor (R), an inductor (L), and a capacitor (C) to passes signals lower frequencies and attenuates signals with higher frequencies.
Here are the formular to form this model(from https://mbe.modelica.university/behavior/equations/electrical/) :

$$V = i_R R$$

$$C\frac{\mathrm{d}V}{\mathrm{d}t} = i_C$$

$$L\frac{\mathrm{d}i_L}{\mathrm{d}t} = (V_b - V)$$

$$i_L = i_R + i_C$$

Where i_R, i_C and i_L are the current on the resistor, capacitor and inductor, V is the voltage on the resistor, C is the capacitance of the capacitor , L is the Inductance of the inductor , t is time and V_b is the source voltage.

In this model , V,i_R, i_C and i_L are the variables we are calculating and plotting , while R,L and C are given values and treated as constants.
What is different from my project proposal is , I didn't extend my model by replacing the resistor to a non-linear resistor(which means the resistance of the resistor changes conditionally), instead I didn't treat my sourse voltage as a constant, but treat V_b as either a constant or a combination of several sine functions of time(which is also plotted). The reason I switched my way to extend my model is that , the impact of non-linear resistor can be decreased by changing to a much-linear resistor, and I realize that the main function of the filter is to filter alternating current sourse and output a signal we want, while my model should show its theoretical result. In fact, the AC sourses did give me a lot of interesting results.

To model this filter, I transformed my equations to :

$$\frac{di_r}{dt} = \frac{i_L - i_r}{CR}$$

$$\frac{di_L}{dt} = \frac{V_b - I_r R}{L}$$

$$I_c = I_L - I_r$$

$$V = I_r R$$

,

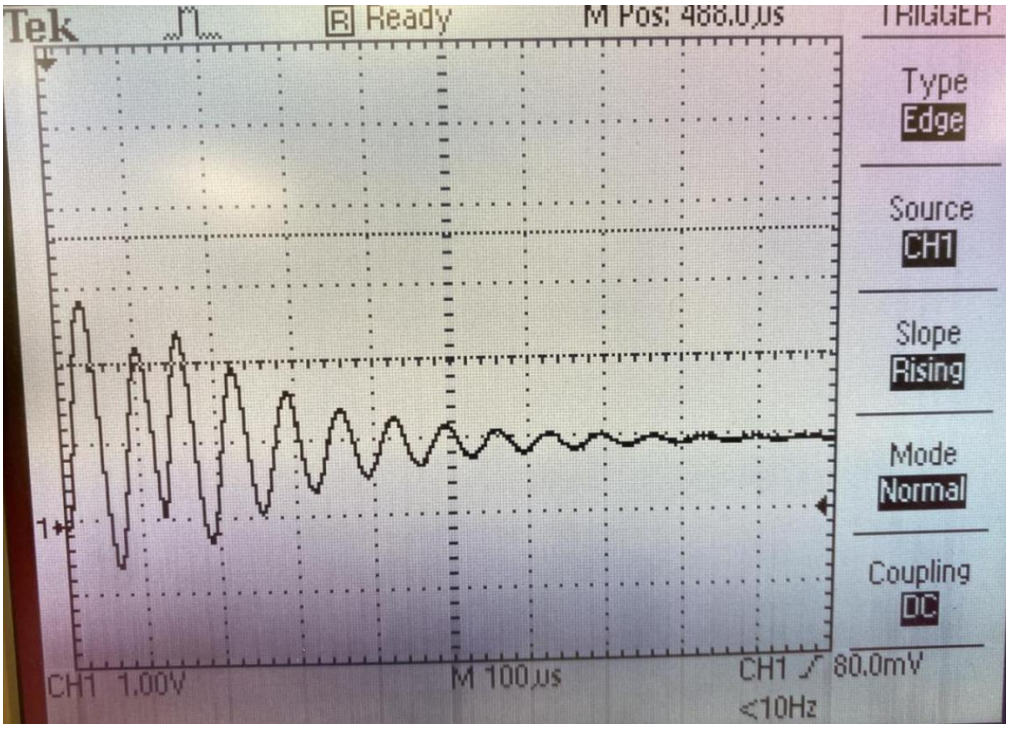
and then used RK4 method to calculate i_R and i_L, then i_c and V are easy to get. Here is how I did it in my code:

```
double dIrdt(double Irr,double ILL,double C,double R){
  double dydt_=(ILL-Irr)/(C*R);
  return dydt_ ;
}

double dILdt(double Vb,double Irr,double R,double L){
  double dydt_=(Vb-Irr*R)/L;
  return dydt_ ;
}



double numerical(double C,double R,double L,double Vb,double dt)
{
  double k1r,k2r,k3r,k4r,k1l,k2l,k3l,k4l;
  k1r=dt*dIrdt(Ir,IL,C,R);

  k1l=dt*dILdt(Vb,Ir,R,L);

  k2r=dt*dIrdt(Ir+0.5*k1r,IL+0.5*k1l,C,R);

  k2l=dt*dILdt(Vb,Ir+0.5*k1r,R,L);

  k3r=dt*dIrdt(Ir+0.5*k2r,IL+0.5*k2l,C,R);

  k3l=dt*dILdt(Vb,Ir+0.5*k2r,R,L);

  k4r=dt*dIrdt(Ir+k3r,IL+k3l,C,R);

  k4l=dt*dILdt(Vb,Ir+k3r,R,L);

  Ir = Ir + (k1r + 2* k2r + 2* k3r + k4r)/6;
  IL = IL + (k1l + 2* k2l + 2* k3l + k4l)/6;

 IC=IL-Ir;
 V=Ir*R;
```

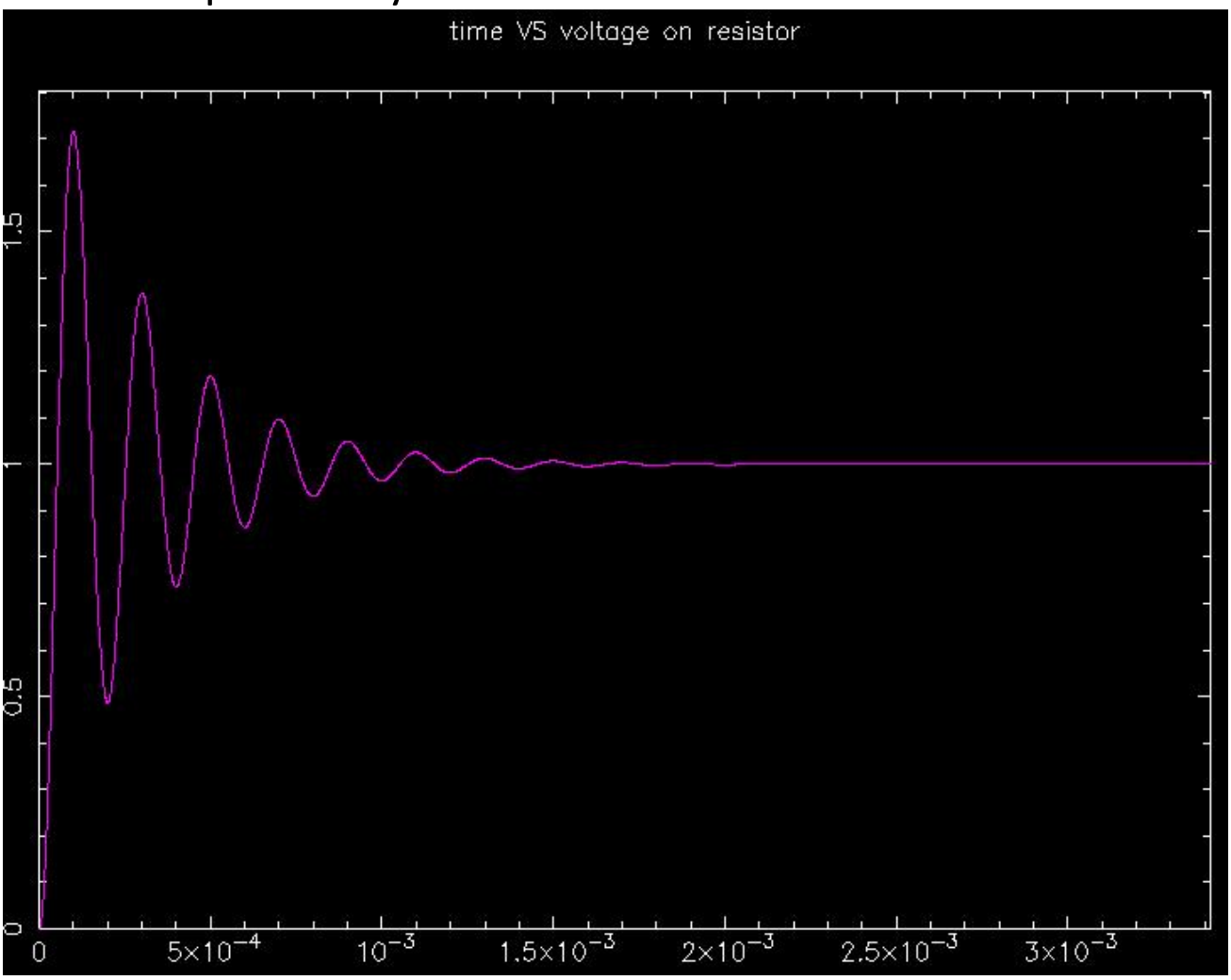
To test my model, with the help from my Physics lab Professor Dr. Daniel FitzGreen(Thank you very , very much ,Dr. FitzGreen! I really appreciate your kindly help!) I did several experienments in lab (using PASCO equipment, an oscilloscope and some addition resistors and capacitors) and got some oscilloscope image for my filter.

(for the oscilloscope image, the virtical axis is the voltage on resistor and the horizontial axis is time, which is exactly the same rule as my plot)
(also, my R,L and C parameters and Vb formulars are the same as the ones we use in experiement)

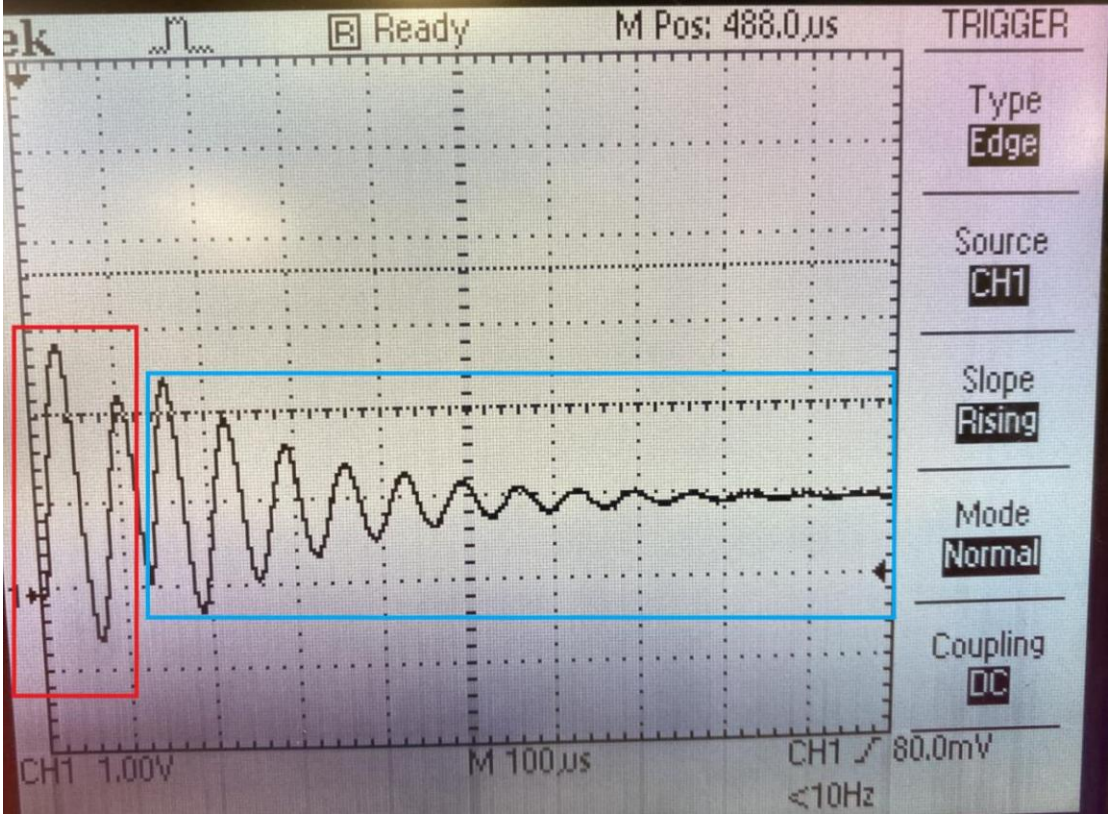
Here is the oscilloscope image for the 1V DC source input(which is case 1 in my model):



while the plot in my model is :



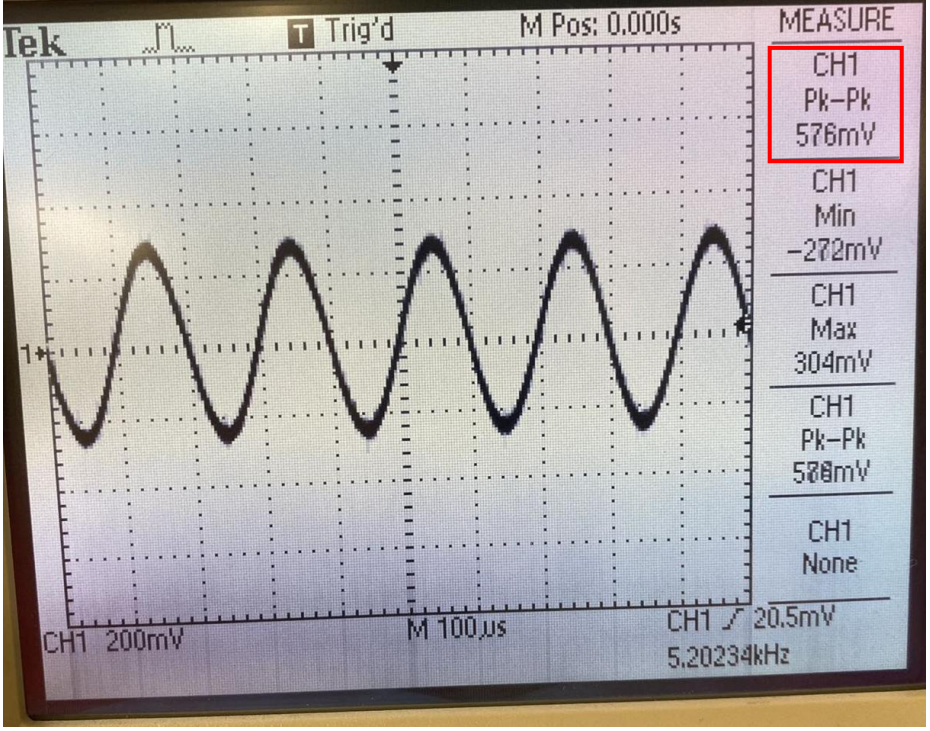
As the plot shows, the voltage across the resistor is a over-damping oscillation graph, while in the experiement,



the blue part looks perfectly similar to the plot, while the red part don't fit it . According to the explaination by Dr. FitzGreen (Thank you again ,Dr. FitzGreen!) , what causes this misfit is that, in my model , the DC sourse I use is a imaginary perfect source, when the program starts, the voltage jumps from 0 to 1V suddenly , while in the real world it has a process to increase from 0 to 1V, which causes this misfit. Also , the oscilloscope is a very sensive instrument and the way it shows the signal is different from my plot, which means there may be a little difference between the image and plot. But in general, they have very similar graphs, so we can consider that they are roughly matching.

To test the behavior of my model at high frequency AC source, we should focus on the amplitude (in the other way to say, the half of peak-to-peak voltage) of the output signal to see how the filter filter out the high-frenquency signal.

Here is my oscilloscope image for 11000 Hz and 1V amplitude AC source input(which is case 4 in my model):



As shows (and marked red) in the image , the amplitude of output voltage is 0.576V.

In my plot, I made it to print out the pk-pk value (for both original source voltage and the filterred output) and in this case it is :

```
the peak-to-peak  voltage of source output is : 2
the peak-to-peak voltage on the resistor is :  0.525534
```

Which shows that the peak-to-peak voltage of the filter in my model is close to the experiement data. Considering the unavoidable experimental error, I think this experiement data can prove that my model works well.

Now it's time to show my result. I made 5 cases to show the property of my model, and 1 additional case to show the interesting fact I found during testing this mode while it will also be my (possible)future work.

Then it's to show my model:

After you run it, it will ask the user to choose the case they want to see:

```
Please choose the type of current sourse you want to model
1. A Direct current sourse
2. A low-frequency sine alternating current sourse
3. A low-frequency alternating current sourse that formed by a few sine waves
4. A high-frequency alternating current sourse that formed by a few sine  waves
5. An alternating current sourse that formed by a few high-frequency and low-frequency sine and cosine waves
6. An interesting additional example of how my filter works like an amlifier to enhance the maximum voltage at a specific frequency
```

The user can choose one of those by entering 1-6, any other input will cause an error (or something that looks like an error) like this :

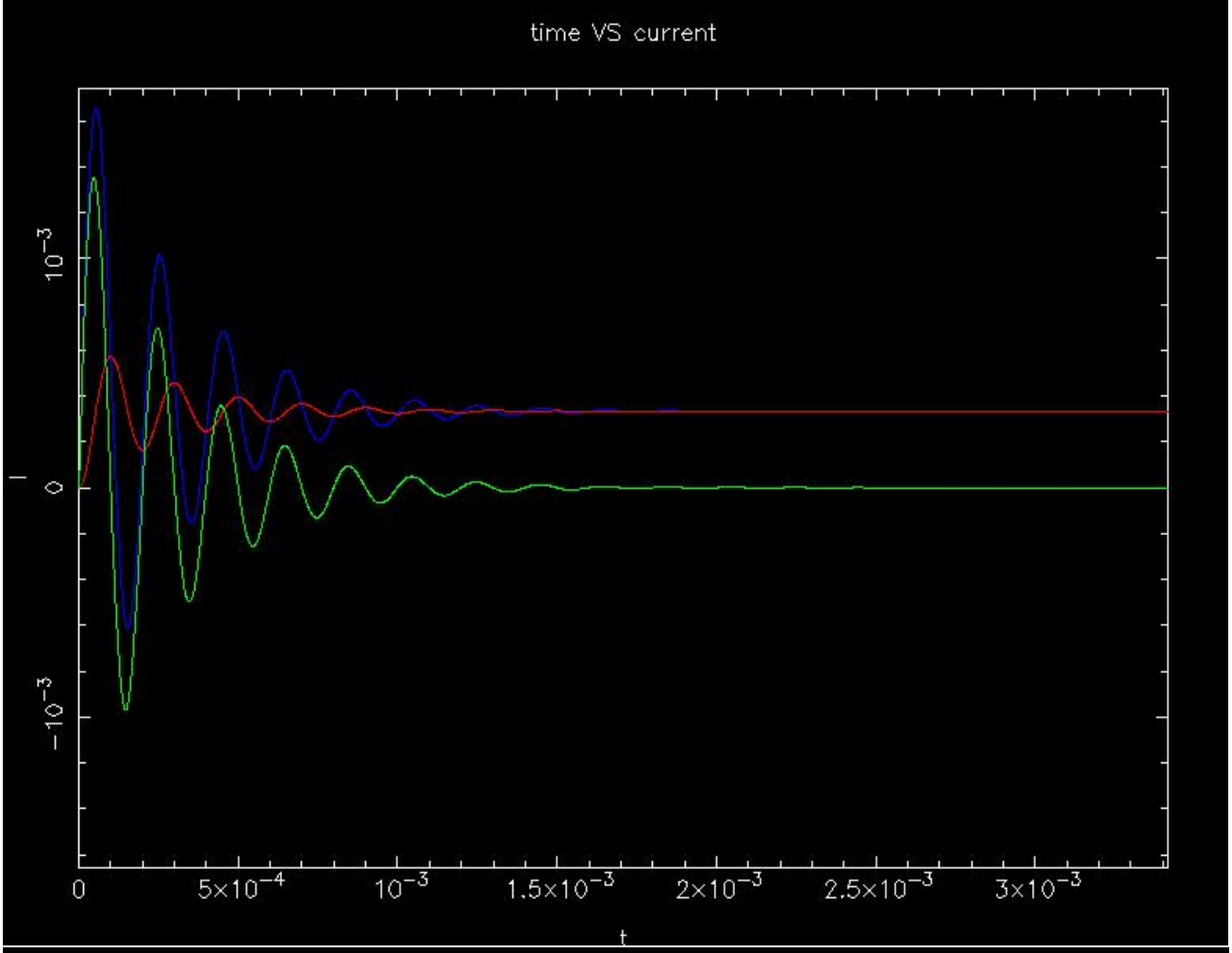
```
Please choose the type of current sourse you want to model
1. A Direct current sourse
2. A low-frequency sine alternating current sourse
3. A low-frequency alternating current sourse that formed by a few sine waves
4. A high-frequency alternating current sourse that formed by a few sine  waves
5. An alternating current sourse that formed by a few high-frequency and low-frequency sine and cosine waves
6. An interesting additional example of how my filter works like an amlifier to enhance the maximum voltage at a specific frequency
7
Error: illegal index
```

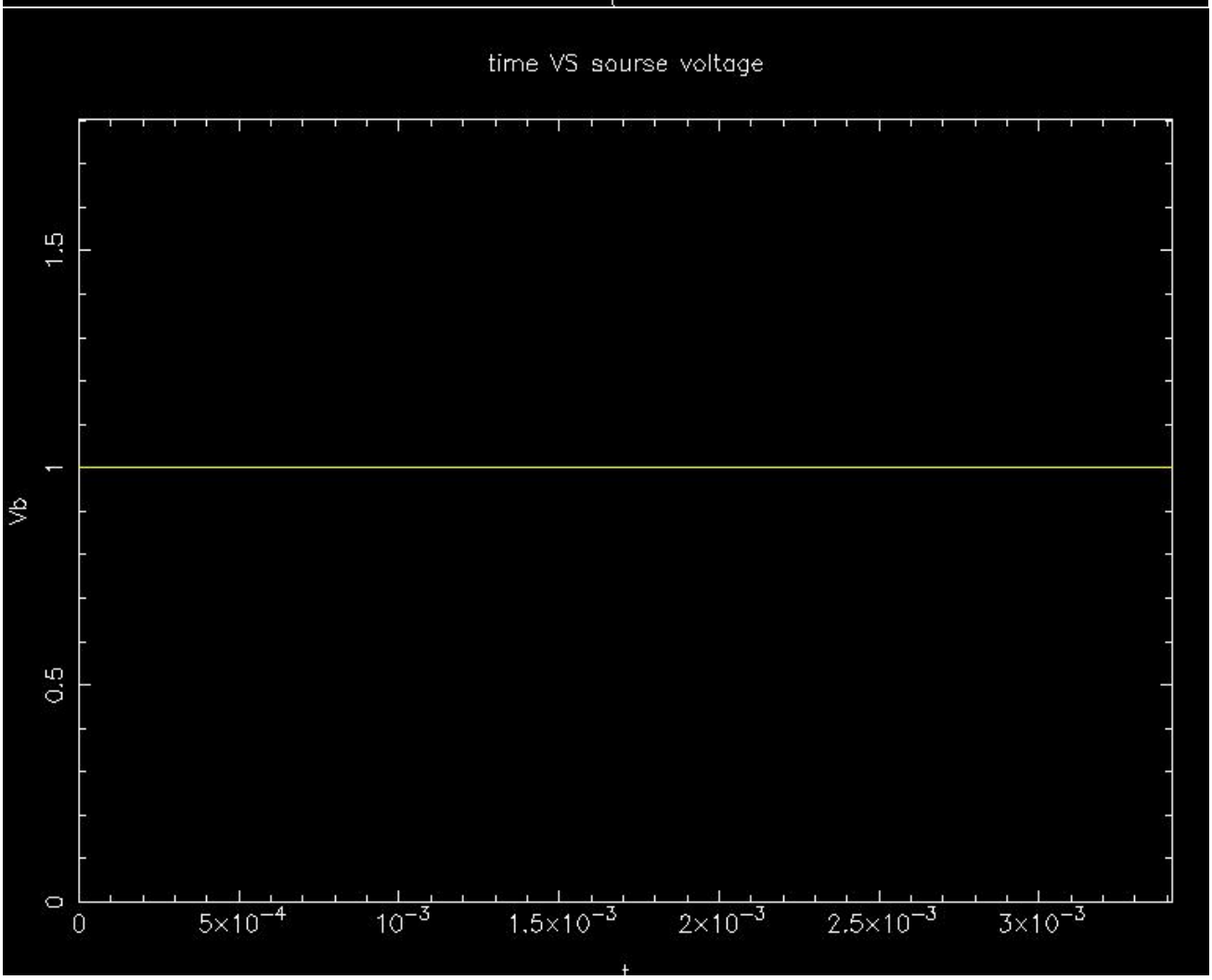
and end the program.

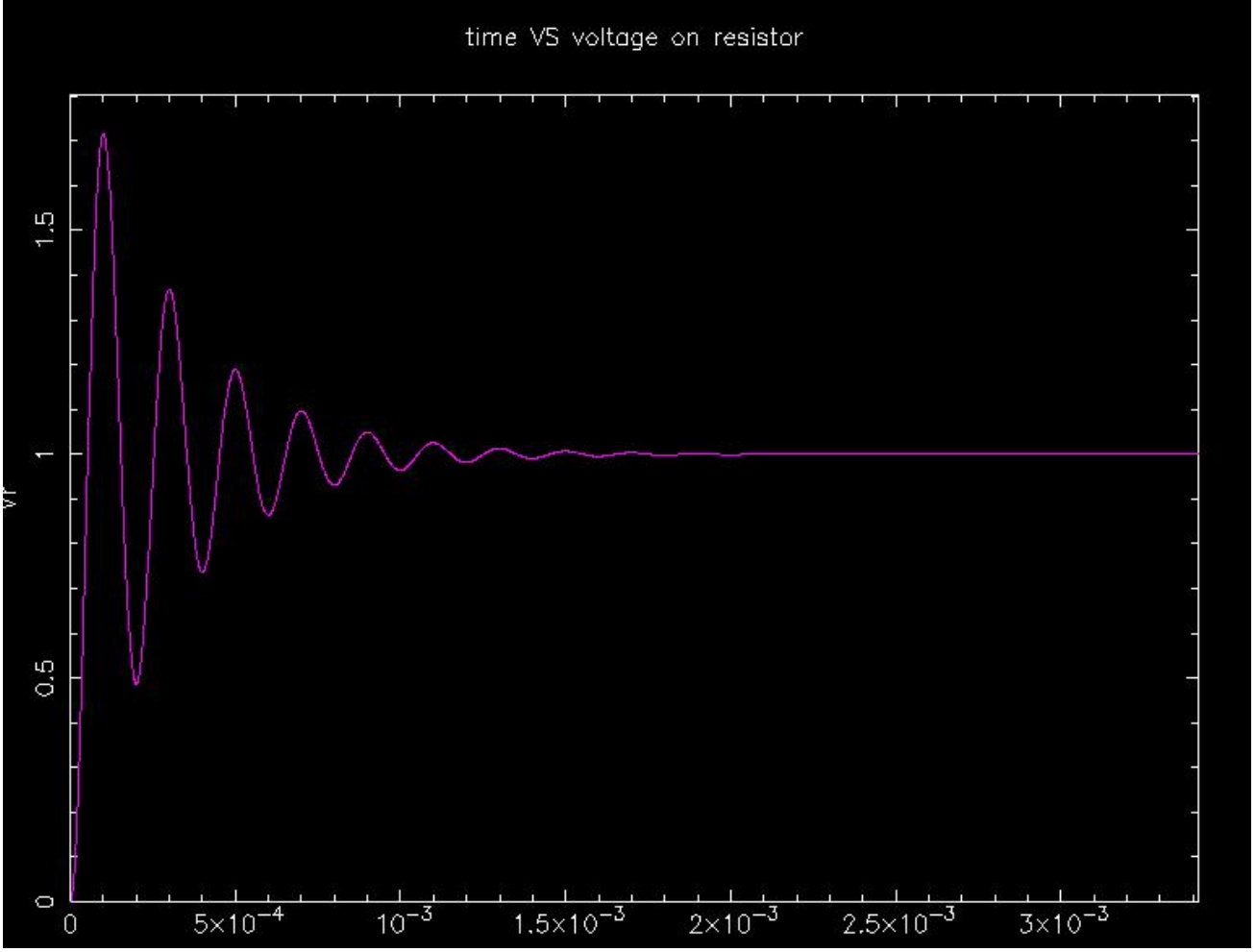
Now choose one of those:

(Rule for all plots: the first graph shows current on each component, which ir is red, iL is blue and iC is green.
the second graph shows sourse voltage.
the third graph shows the voltage on the resistor, which is also the result we want to get.)

1.A Direct current sourse

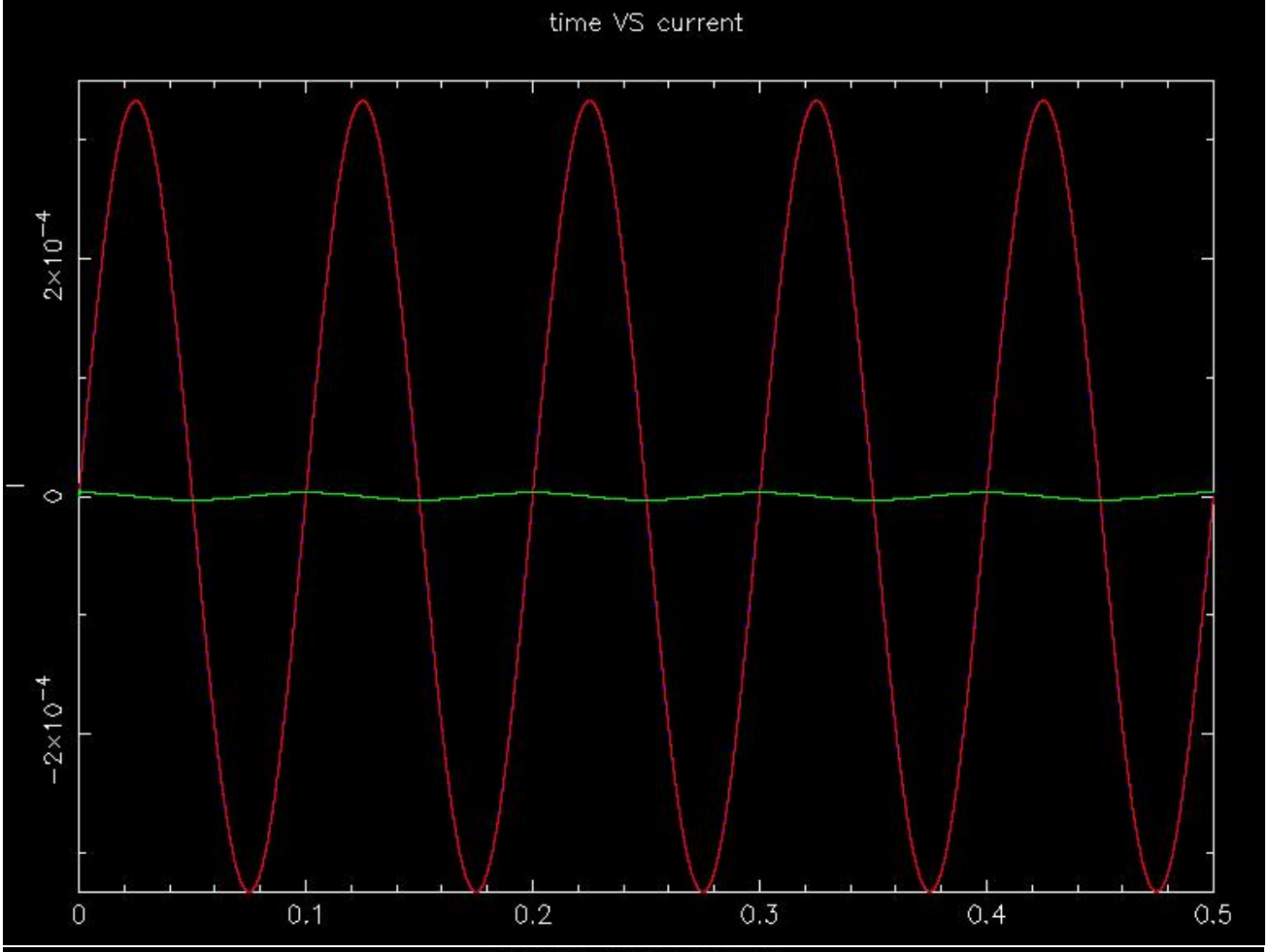

time VS current
I
10^-3
0
-10^-3
0
5×10^-4
10^-3
1.5×10^-3
2×10^-3
2.5×10^-3
3×10^-3
t

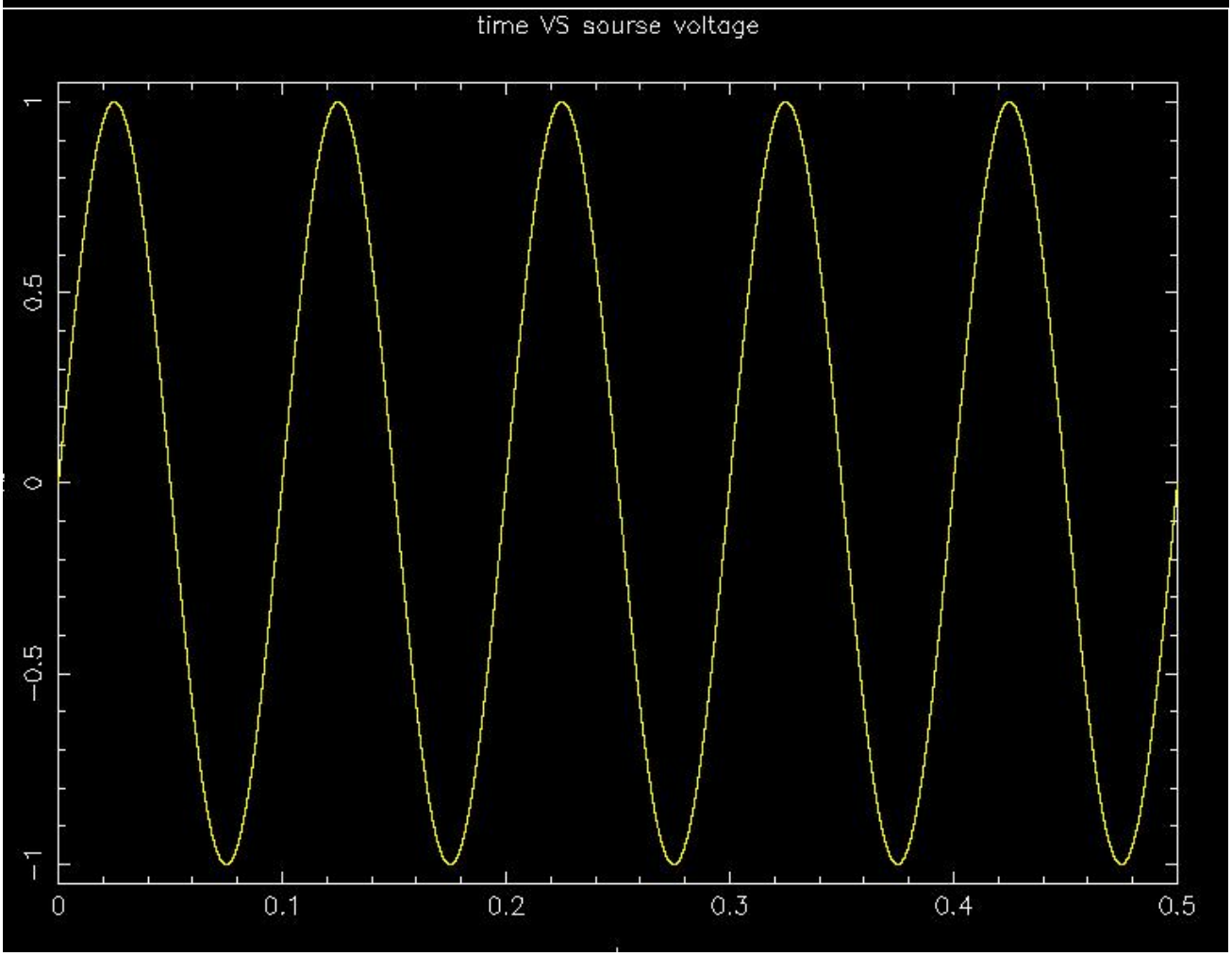

time VS sourse voltage
Vb
1.5
1
0.5
0
0
5×10^-4
10^-3
1.5×10^-3
2×10^-3
2.5×10^-3
3×10^-3
t



These are the plots it will show.
This case show the impact of filter on DC source in a over-damped situation, which is explained in the test section.
In this case, pk-pk value is meaningless so it isn't shown.

2. A low-frequency sine alternating current sourse

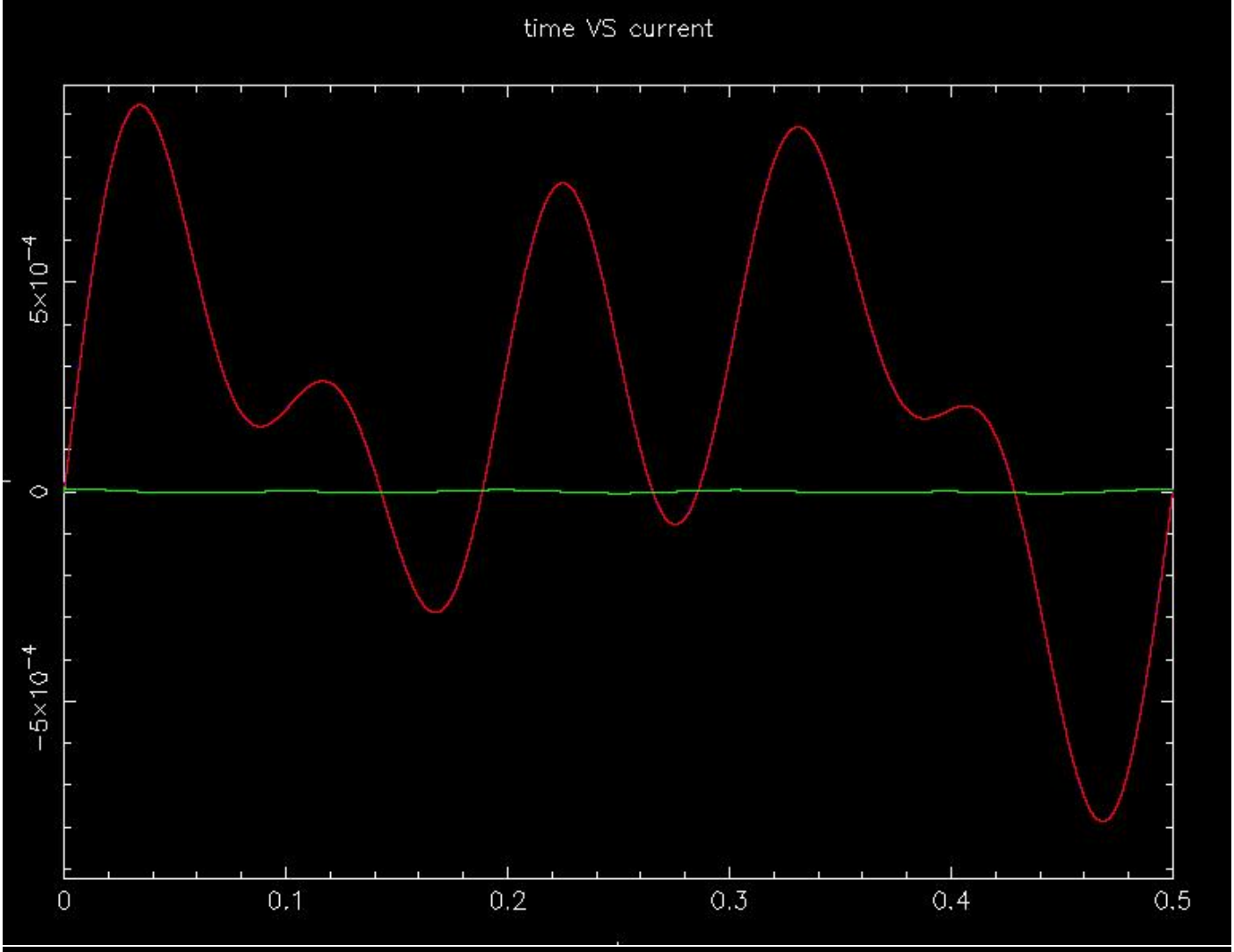

time VS current
I
2×10⁻⁴
0
−2×10⁻⁴
0
0.1
0.2
0.3
0.4
0.5

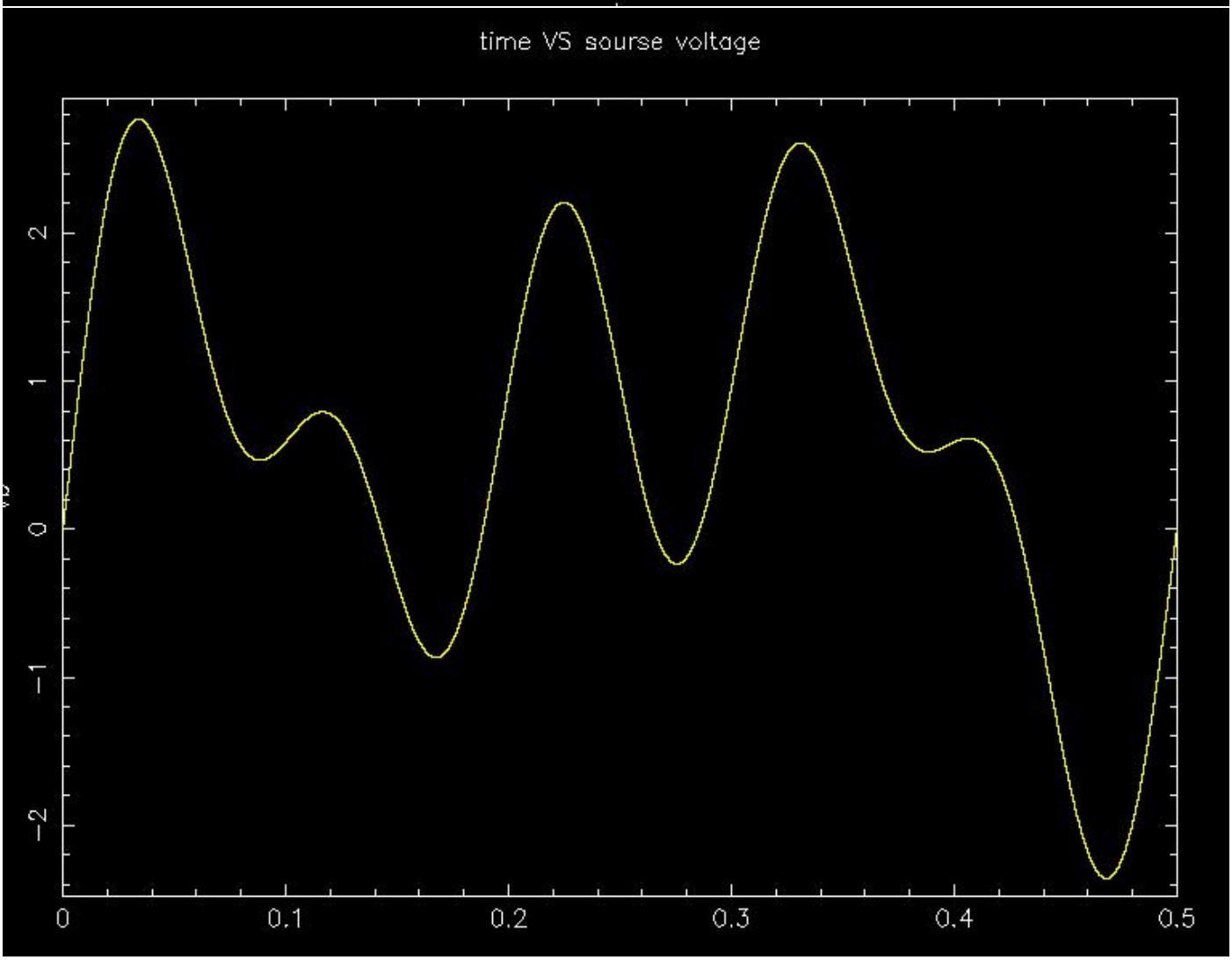

time VS sourse voltage
1
0.5
0
−0.5
−1
0
0.1
0.2
0.3
0.4
0.5

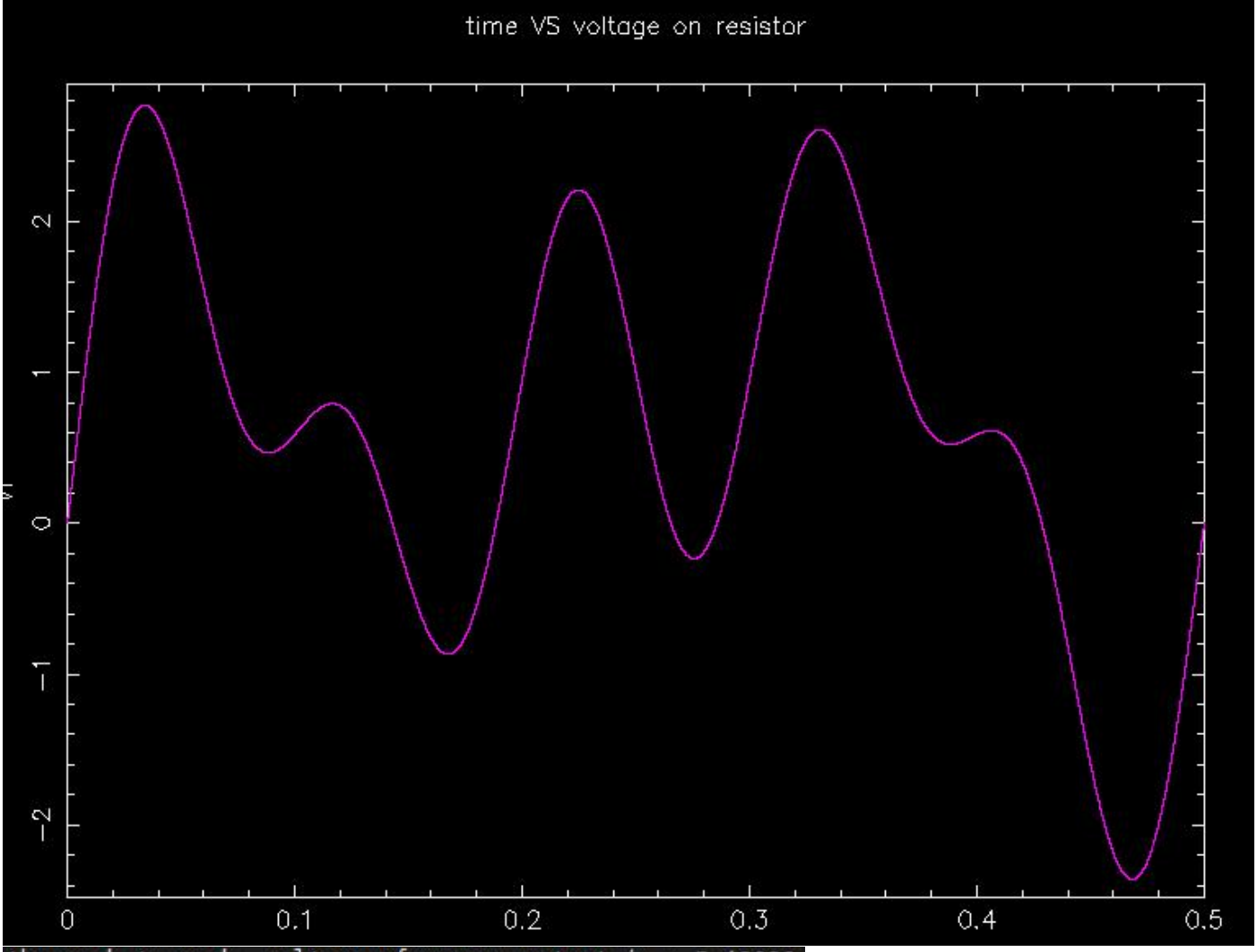
```
the peak-to-peak  voltage of source output is : 2
the peak-to-peak voltage on the resistor is :  2.00001
```

In this case , we can see the input and output signal is almost exactly the same and so on to its amplitude, which shows that our filter allows low-frequency signal passes perfectly.

3. A low-frequency alternating current sourse that formed by a few sine waves


time VS current
5×10⁻⁴
0
−5×10⁻⁴
0
0.1
0.2
0.3
0.4
0.5


time VS sourse voltage
2
1
0
−1
−2
0
0.1
0.2
0.3
0.4
0.5




```
the peak-to-peak  voltage of source output is : 5.12926
the peak-to-peak voltage on the resistor is :  5.12927
```

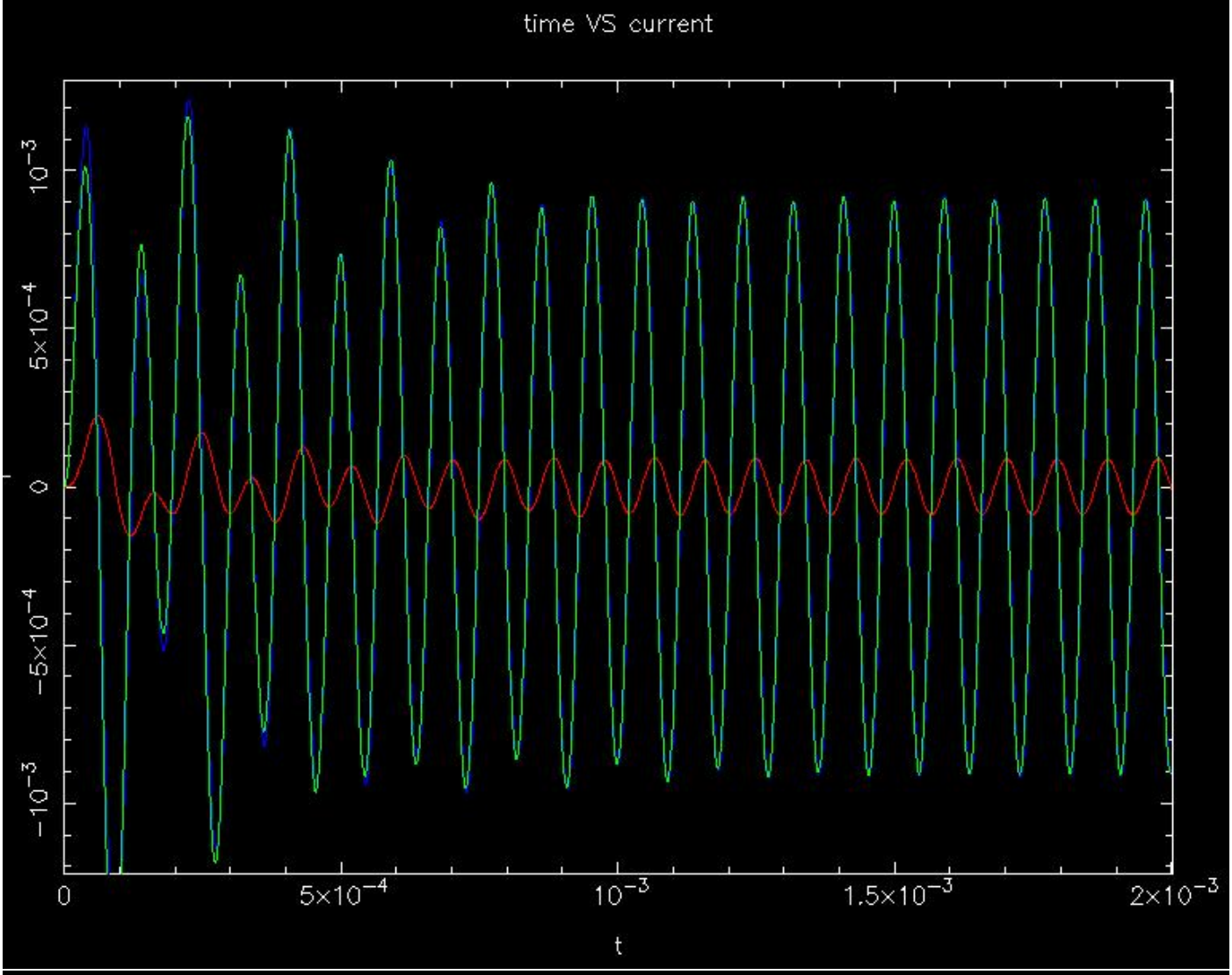
This AC sourse is formed by 4 different sine waves :

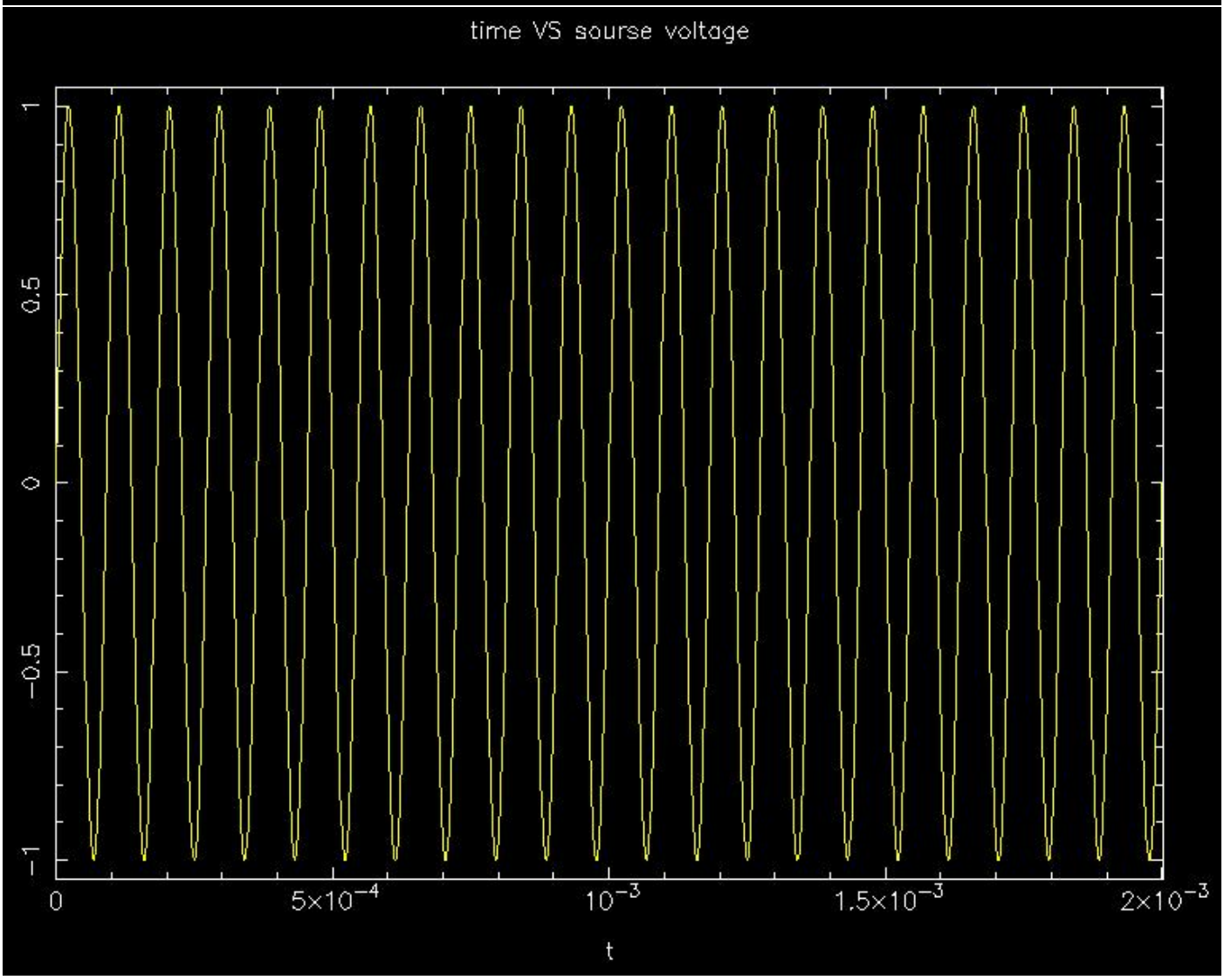
```
double Vb3(double t1){double res=sin(2*M_PI*t1)+sin(4*2*M_PI*t1)+sin(10*2*M_PI*t1)+sin(6*2*M_PI*t1) ;return res ;}
```

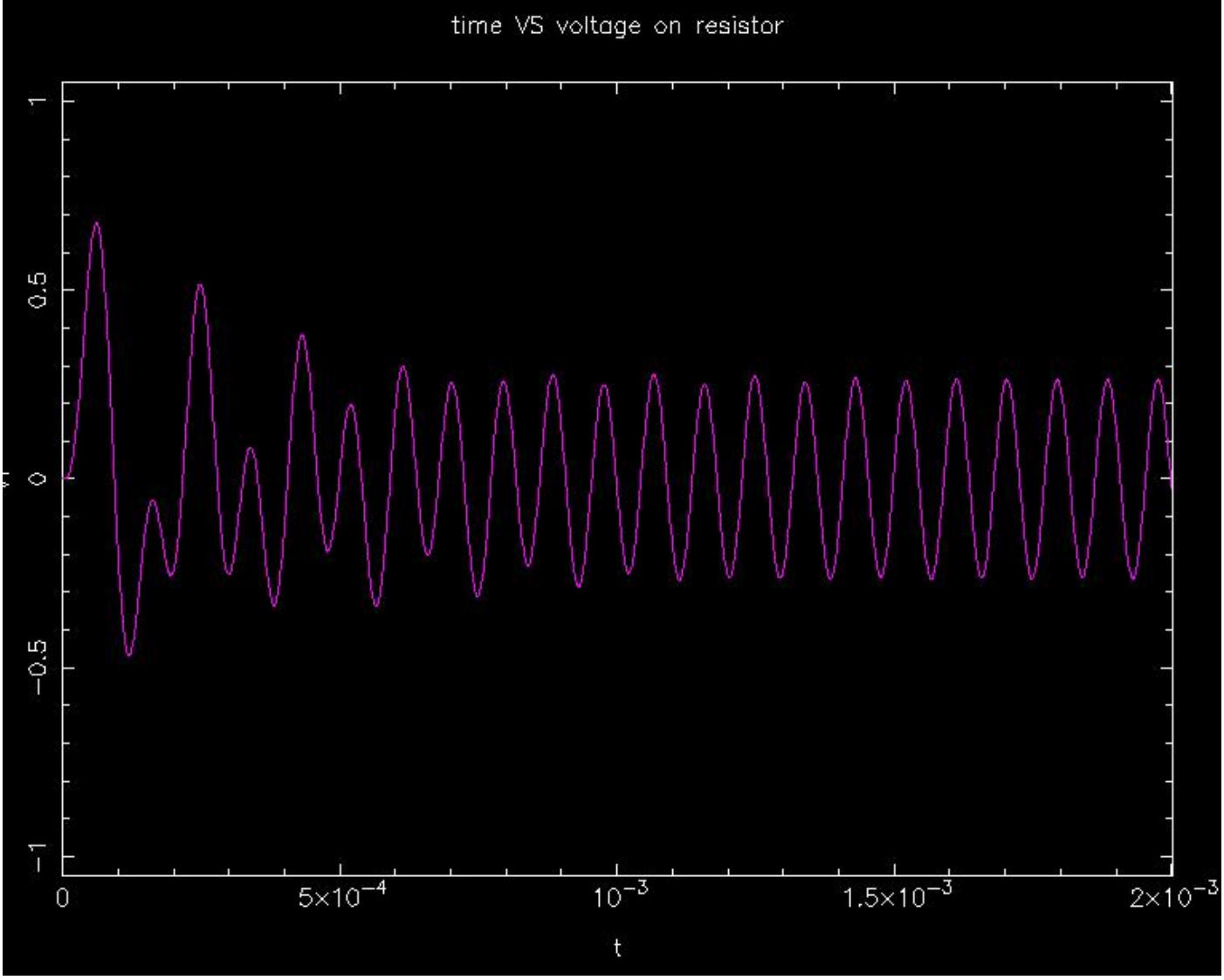
which shows the same conclution as the pevious one:signal passes perfectly, no amplitude changes.
Also, its obvious that these two graph shows in the low-frequency cass, the current on L and R are almost the same, which means(and shown in plots) iC is very close to 0, and that fits the property of capacitor: blocks low frequency signal and allows high ones passing.

This case is to compare case 5.

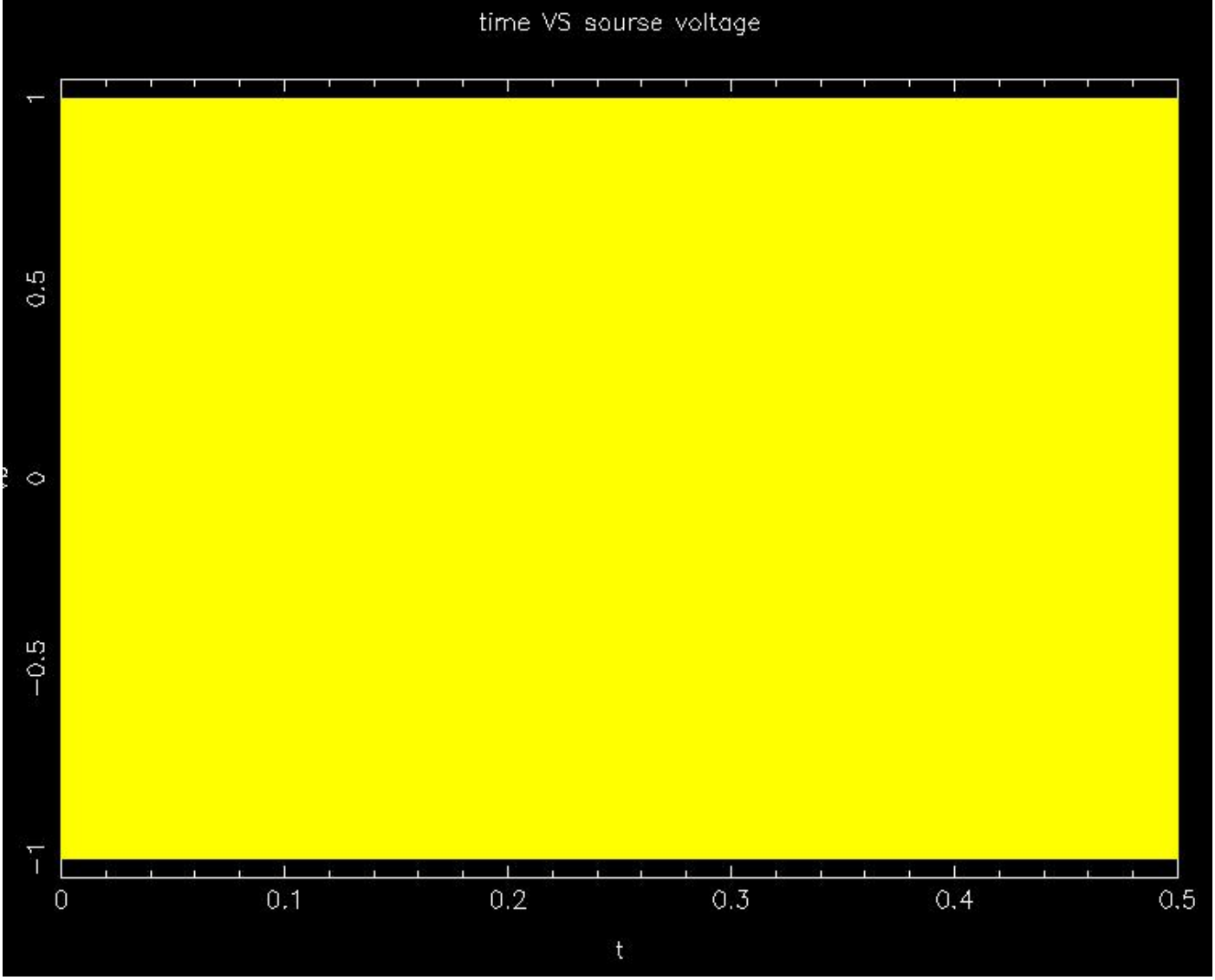
4. A high-frequency alternating current sourse that formed by a few sine waves

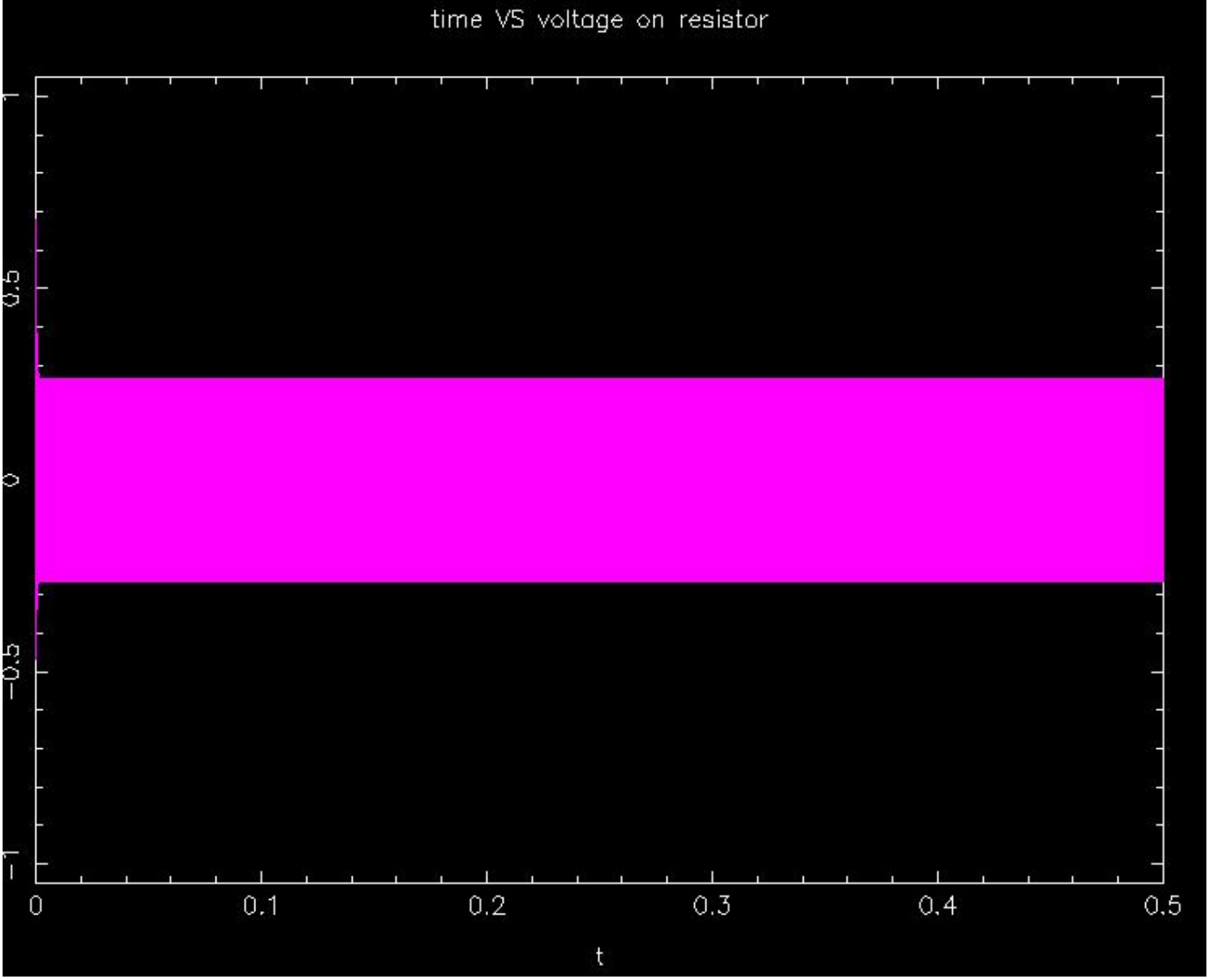

time VS current
10^-3
5×10^-4
0
-5×10^-4
-10^-3
0
5×10^-4
10^-3
1.5×10^-3
2×10^-3
t


time VS sourse voltage
1
0.5
0
-0.5
-1
0
5×10^-4
10^-3
1.5×10^-3
2×10^-3
t



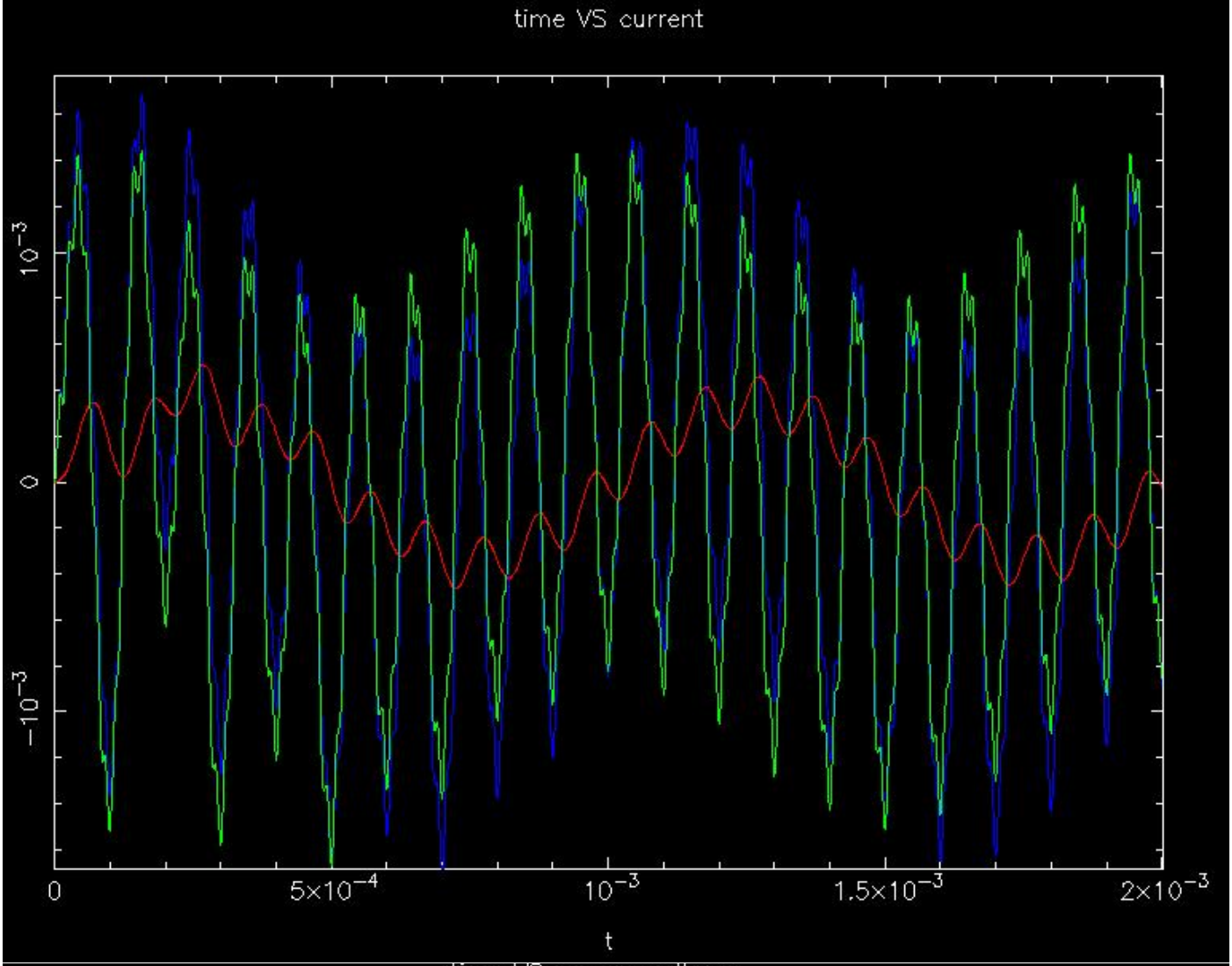
(Let’s use the abbreviation “frqcy” instead of “frequency”)
As this plot shows , we can see the the amplitude of the high-frqcy (which is 11000Hz) signal is weakened a lot after being filterred ,which matches the experienment image.
However, in this case and case No.5, the plot didn’t wait util the filterring is over because if we do so, the entire graph will be extremely messy that we can hardly get any useful infomation from them except the voltage amplitude (which shows pretty good however) due to the too-high frqcy(which looks like:

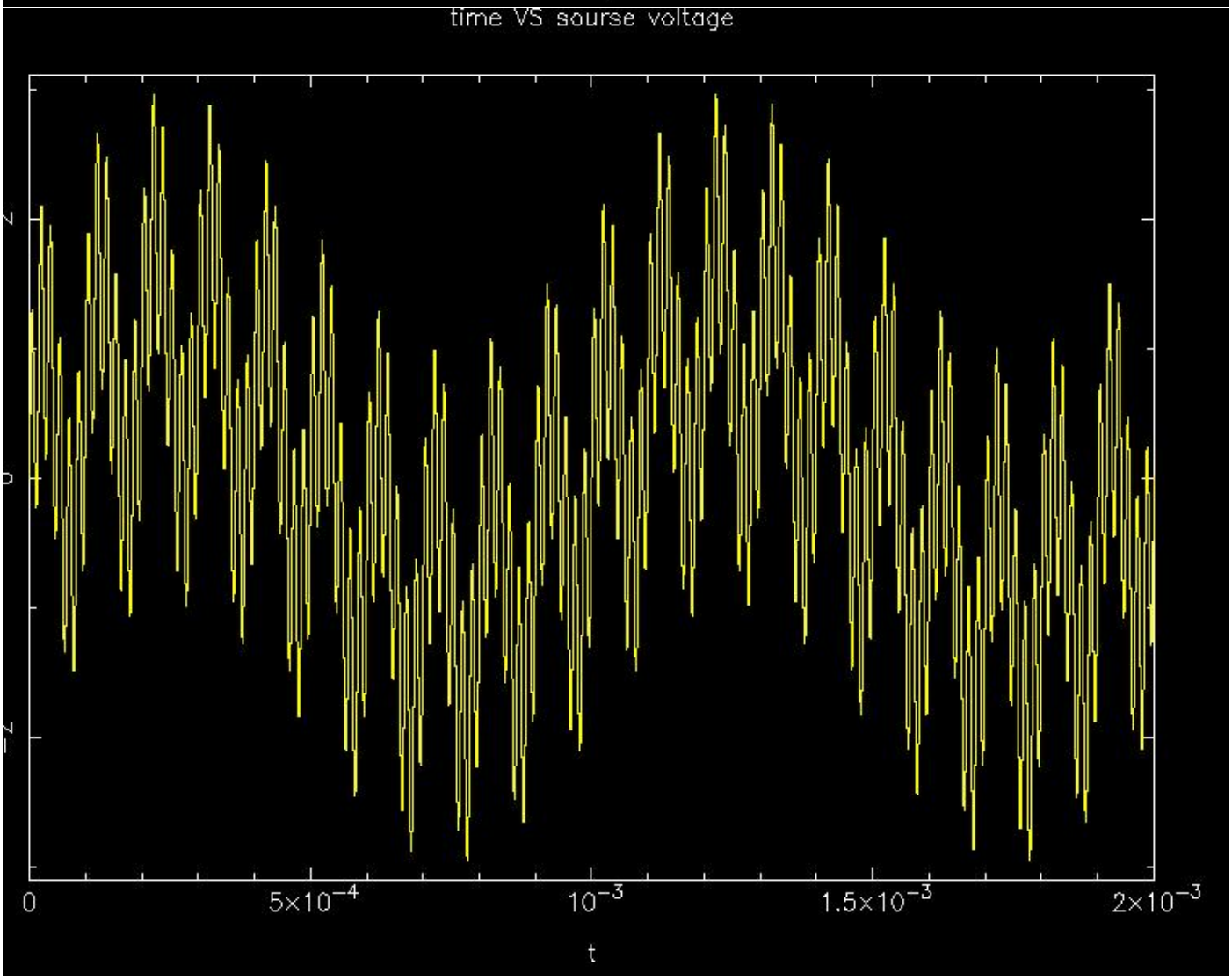

time VS sourse voltage
1
0.5
0
−0.5
−1
0
0.1
0.2
0.3
0.4
0.5
t

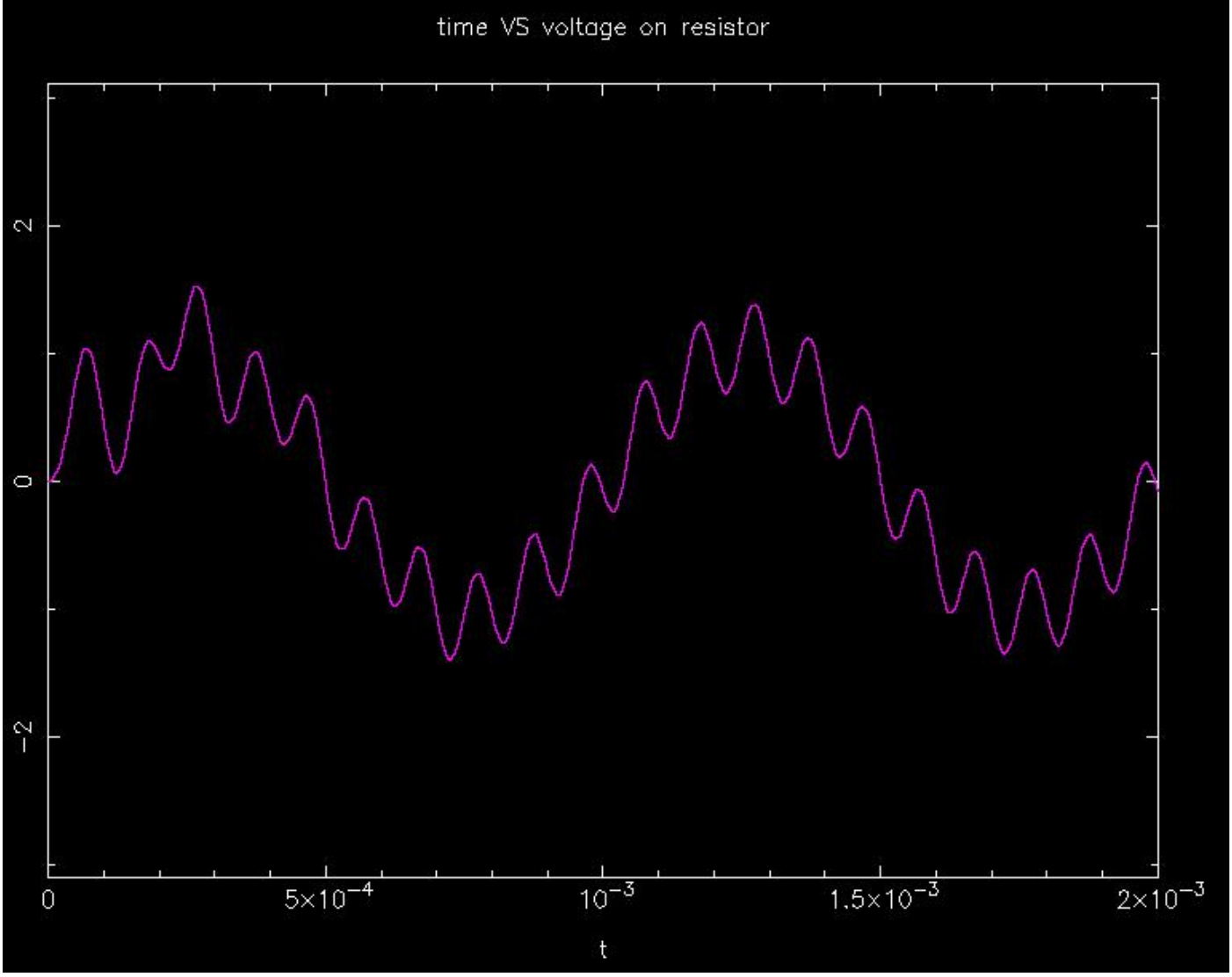


So that I cutted the time interval and ask it to just calculate the first few perieds , but this makes the pk-pk value untrustable, so if the user want to see its amplitude, they have to change the time value (or set it as one of my future work).

5. An alternating current sourse that formed by a few high-frequency and low-frequency sine and cosine waves


time VS current
10^-3
0
-10^-3
0
5×10^-4
10^-3
1.5×10^-3
2×10^-3
t


time VS sourse voltage
0
5×10^-4
10^-3
1.5×10^-3
2×10^-3
t



this oruginal signal is formed by 4 sine functions:

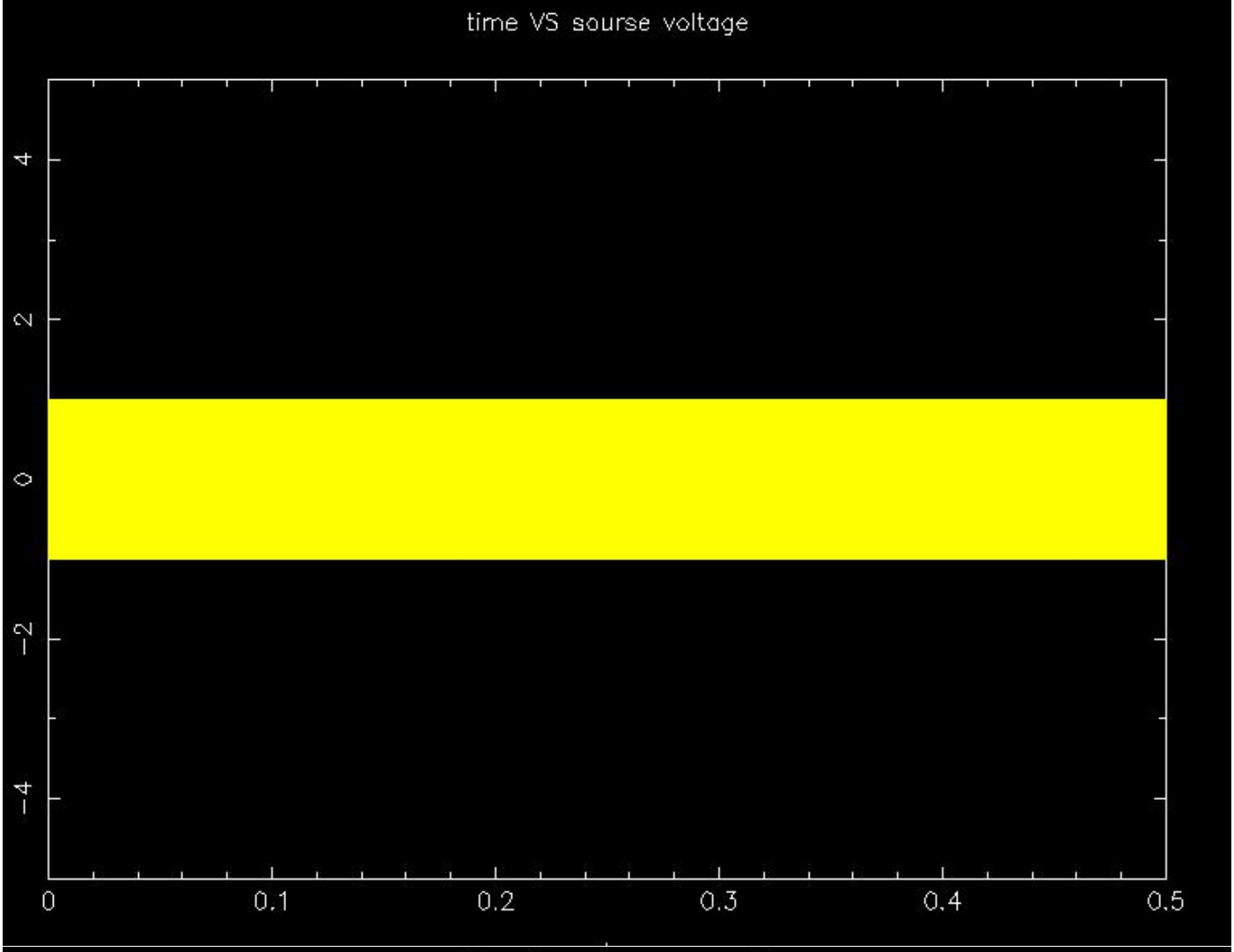
```
double Vb5(double t1){double res=sin(2*M_PI*t1)+sin(10000*2*M_PI*t1)+sin(1000*2*M_PI*t1)+sin(60000*2*M_PI*t1) ;return res ;}
```

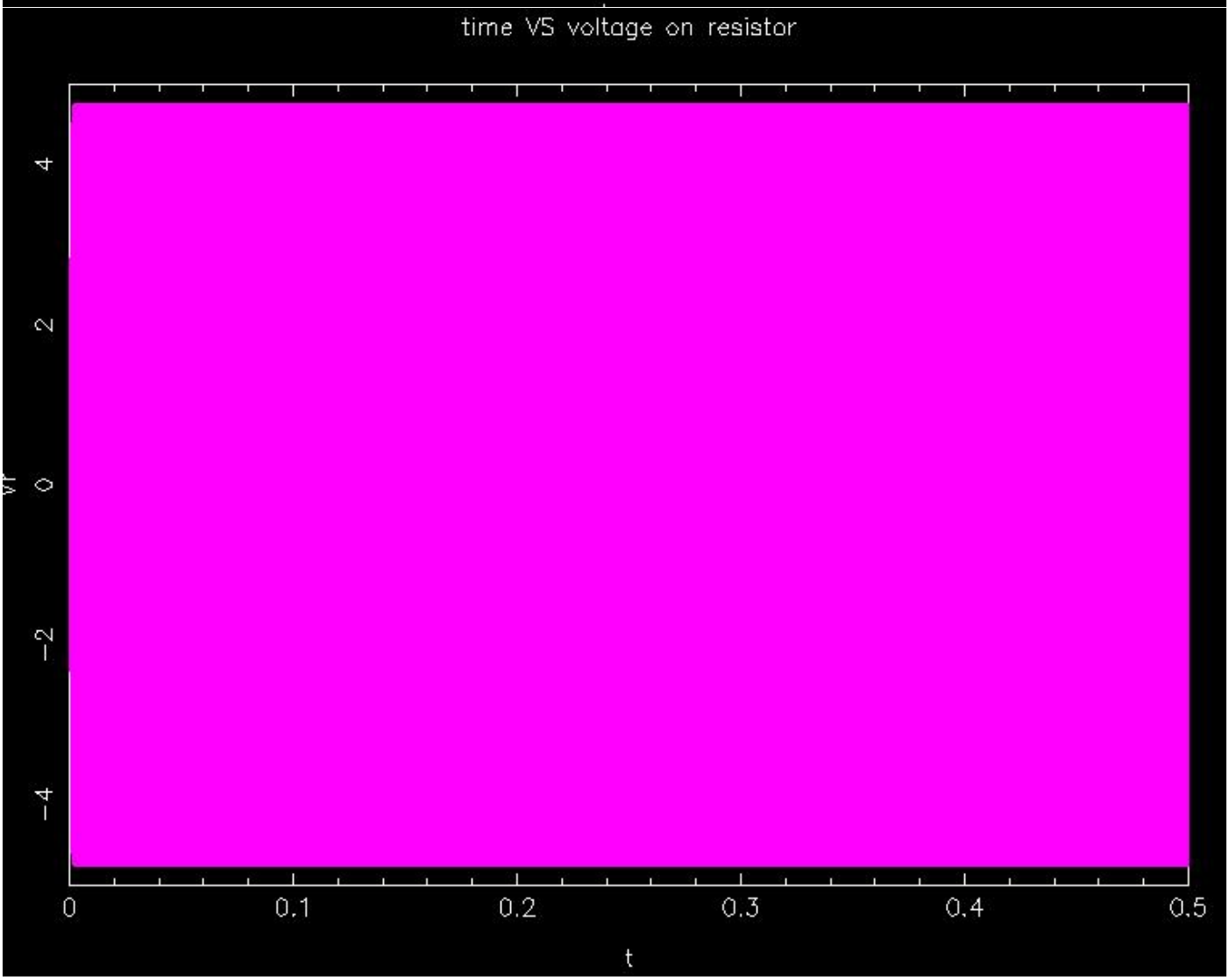
which contains one low-frqcy function, one mid-frqcy one and 2 high-frqcy ones.
We can see in the plots, the input signal was very noisy, but after the filter works , it becomes nice and pretty.

This case is , actually , showing the real usage of this filter. In the real word, AC voltage don't just contain one frequency, imstead they are conbinations of many sine waves with different frqcys , and it makes them become very noisy as the input signal. So that, this is the main goal of my filter: doing noise reduction, filterring out the unneeded noisey high-frqcy

wave by weakening them until they are too weak to influence the signal graph and leaving the nice low-frqcy wave we want. In this case, the sine wave that with frqcy of 1Hz and 1000Hz are allowed to pass, while the 10000Hz and 60000Hz ones are filterred out. Compare to case 3, we can see the signal changes a lot in this case , while in case 3 it is almost unchanged.
It also shows the low-pass property of this filter.

Finally , since this model fits experiement data good and it shows the low-pass property and main usage of the filter perfectly, this model is justified.
For the future work, I found a interesting property of this filter: when the input frqcy is in a certain range, the filter works like an amlifier to enhance the maximum voltage .
As shown in case 6, when the input frqcy is 5000Hz, the comparison of input and output plot is:


time VS sourse voltage
4
2
0
−2
−4
0
0.1
0.2
0.3
0.4
0.5


time VS voltage on resistor
4
2
0
−2
−4
0
0.1
0.2
0.3
0.4
0.5
t

```
the peak-to-peak  voltage of source output is : 2
the peak-to-peak voltage on the resistor is :  9.52937
```

which looks unblieveable but actually makes sence, since that related to the property of inductor: when it is discharging, it output energy by applying excess voltage to the circuit, hence it can enhance the maximum voltage of a circuit, which means when the frqcy is in a certain range(which I blieve is related to the parametors of the components), it can behaive like an amplifier to amplify the maximum voltage.
Therefore, my futurework will be testing this property, looking for the best enhanced frqcy and plot out the relationship between frqcy and maximum voltage.

That’s all for my project. Thank you so much for reading!